

Classification of Motor Imagery Hands Movement Using Levenberg-Marquardt Algorithm based on Statistical Features of EEG Signal

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Abstract—Motor Imagery (MI) is a highly supervised method nowadays for the disabled patients to give them hope. This paper proposes a differentiation method between imagery left and right hands movement using Daubechies wavelet of Discrete Wavelet Transform (DWT) and Levenberg-Marquardt back propagation training algorithm of Neural Network (NN). DWT decomposes the raw EEG data to extract significant features that provide feature vectors precisely. Levenberg-Marquardt Algorithm (LMA) based neural network uses feature vectors as input for classification of the two class data and outcomes overall classification accuracy of 92%. Previously various features and methods used but this recommended method exemplifies that statistical features provide better accuracy for EEG classification. Variation among features indicates differences between neural activities of two brain hemispheres due to two imagery hands movement. Results from the classifier are used to interface human brain with machine for better performance that requires high precision and accuracy scheme.

Keywords— *Motor imagery (MI), discrete wavelet transform (DWT), brain machine interface (BMI), neural network (NN).*

I. INTRODUCTION

Patients with severe neuromuscular disorders and paralyzed from higher level spinal cord injury or last stage amyotrophic lateral sclerosis (ALS) patients are not capable of moving any voluntary muscle [1]. Their sensory and cognitive functions become only way to communicative with surroundings. Such patients can be motivated, trained for MI movement to execute real movement by incorporating with Brain Computer Interface (BCI) utilizing neural activity to do like cursor control, selection of letters or words, control of prosthesis, navigation of wheelchair etc.

Imagery movement of two hands can be distinguished according to the change of the electrical (neural) activity. MI is the procedure where imagining motor activity human observe activation of motor cortex with or without external trigger. Motor imagery movements mimic original motor act without even accomplishing any real motor activity. Motor performance, sensory provocation, and mental imagination modify the functional activity of the cortex and results in increasing or decreasing amplitude. Motor imagery

movements dramatically change the neural activity of the primary sensorimotor areas in a quite indifferent way as real performed movements [2].

Practical implementation of mind machine interface (MMI), brain machine interface (BMI) or direct neural interface (DNI) system requires a well organize brain signal processing method to extract responsible features and to perform classifications with better accuracy. Several methods have been reported to extract features in different domains where wavelet transformation (WT) based features extraction method is highly efficacious method, because it works greater with the random characteristics of EEG signals than other scheme. Wavelet-based features like wavelet entropy, wavelet coefficients and wavelet statistical features (mean, median, and standard deviations) most commonly used for normal EEG analysis in clinical applications as well as in research areas also [3].

In literature, two new features, (1) multifractal cumulants and (2) predictive complexity measure are introduced in addition with band power feature [4]. Linear Discriminant Analysis (LDA) classifier is used to classify and compares classification accuracy with each other (features). A multilayer perceptron neural network (MLPNN) based classification model is [3] proposed using EEG signal. DWT and MLPNN outcomes classification accuracy for five different experiments are above 95% for individuals. Authors in [5] proposed a scheme to calculate relative wavelet energy in terms of approximation coefficients and detailed coefficients. Extracted features are given as input to the SVM, K-nearest neighbor (K-NN), multi-layer perceptron (MLP) and Naive Bayes classifiers with classification accuracy 98.75%, 98.21%, 98.21%, 83.57% respectively. In [6] authors unite time and frequency features approach for the purpose of EEG signals differentiation between epileptic and healthy subject. Time domain and frequency domain features are extracted using the cross correlation (CC) and power spectral density (PSD) method respectively. The combination of the cross correlation and power spectral density method for EEG classification is very effectual. Stockwell Transform (ST) dependent study of EEG signals are introduced during performing different mental tasks or assignment [1].

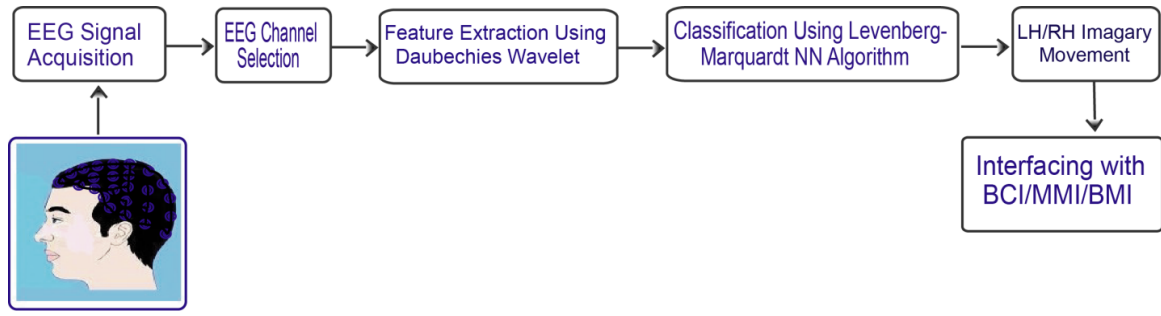


Fig. 1. Block diagram of the classification of imagery LH/RH movement.

kNN, LDA and SVM classifiers were employed to test the classification strength of the proposed features. Accuracy achieved to distinguish the multi-class problems (five mental tasks) is ranging between 84.72% and 98.95%. Wavelet pack entropy features are selected to classify a baseline task from a cognitive task with SVM classifier which provides overall 87.5–93 % accuracy [7]. This study has used a database where seven subjects performed five assign tasks—baseline (open eyes) task, multiplication, visual counting, mental letter composing, and geometric object rotation task.

Many wavelet based research work accomplished [5] except using wavelet statistical features. This paper proposes that wavelet statistical features can be used to classify EEG signal in addition with better classification accuracy. This paper also investigate the performance of the Levenberg-Marquardt algorithm to classify right and left hands imagery movement using Daubechies 4 wavelet.

This paper is organized as follows: Section II illustrates materials and proposed method. Section III states the results and discussion is in section IV. Finally section V concludes the paper.

II. MATERIALS & PROPOSED METHOD

A. EEG Data Acquisition & Description of Dataset

This dataset is collected from internet [4] which is taken at INRIA Renne-Bretagne Atlantique using the OpenVIBE, an open-source and free software [4]. This data set contains EEG signals of one subject performing left hand and right hand motor imagery. Five hundred sixty trials of motor imagery (280 trials per class) were captured over a 2 week period.

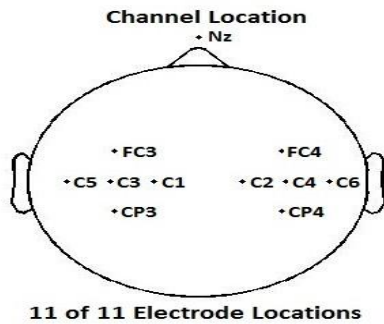


Fig. 2. Electrode placement system of the subject.

During data acquisition same experimental protocol is used as Graz BCI protocol [2], [4] to differentiate the two mental conditions. The experimental task is to imagine the movement of one hand (left/right) for specific time of each trial. Each trial lasts for 8 seconds including resting, imagery movements of either left or right hand and for feedback. Sampling rate of EEG data is 512 Hz and recorded from frontal, central and central frontal lobe using the following 11 electrodes: C1, C2, C3, C4, C5, C6, CP3, CP4, FC3, FC4 and a nose reference electrode Nz [4]. Such electrodes placement helps to employ a discrete Laplacian spatial filter [8] over C3 and C4 to collect better noise free signal, as recommended in [4]. All electrodes placement are depicted in Fig. 2, Nz is the reference electrode. Six electrodes from ear to ear band are very crucial for MI.

B. Channel Selection and Experimental Flowchart

Eleven channels have been used including reference electrode for data collection but our interested channels are above the sensorimotor cortex region. The motor imagery EEG signal is predominant in sensorimotor cortex, corresponding to electrode positions C3 and C4 [4]. In our dataset each electrode collects data of both L/R hands data. Based on the universal truth that left hemisphere controls our right side of the body and right hemisphere powers over left part of the body, we selected C3 channel data for right hand imagery movement and C4 channel data for left hand imagery movement. Figure 1 represents the complete work flow starting with taking data from subject and after classification of L/R hand; output result can be used to interface BCI or BMI.

C. Discrete Wavelet Transform (DWT)

DWT is a novel tool for decomposition and reconstruction filters to extract non stationary characteristics of EEG signal. In this work db4 wavelet is used to analyze signals into 5 levels of sub bands frequencies using continuous high pass and low pass filtering shown in Fig. 3. The high-pass filter, g is the discrete mother wavelet and h is the mirror version of low-pass filter. DWT decomposes a provided signal into increasingly finer detail depending on two sets of fundamental functions [9], the wavelets and the scaling functions of a signal $S(t)$ as follows:

$$S(t) = \sum_k 2^{j_0/2} a_{j_0}(k) \phi(2^{j_0 t - k}) + \sum 2^{j/2} d_j(k) \psi(2^j t - k) \quad (1)$$

Where $\phi(t)$ and $\psi(t)$ are the basic scaling and mother wavelet functions respectively.

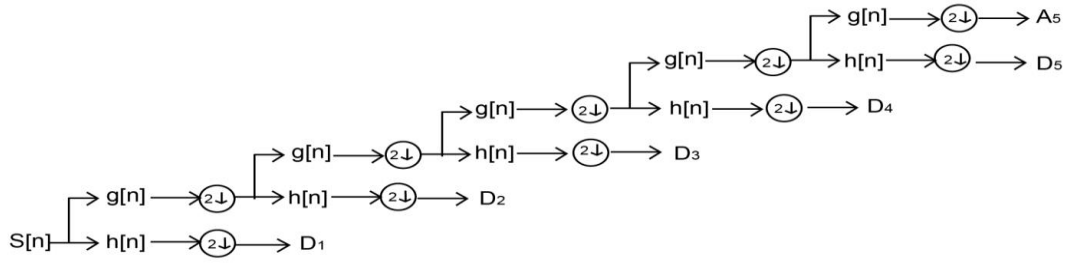


Fig. 3. Decomposition of EEG segmented signal using 5th order Daubechies wavelet.

The sampled output signals from the first filters are called first level approximation, A_1 and detail coefficients, D_1 of the original signal. Approximations and details are the high-scale, low-frequency components and the low-scale, high-frequency components of the signal respectively. The two filters are related to each other and they are defined as a quadrature mirror filter. Approximation and detail coefficients of subsequent level are estimated by using the approximation coefficient of the earlier level. The required decomposition level depends on the commanding frequency components of the signals [3]. After decomposition significant 10 features are collected to make feature vector for further processing to classify imagery LH/RH movement.

D. Neural Network (NN)

In ANNs, the intelligence based on the interrelation of weights between neurons to neurons used in the NN design. Therefore, training course of neurons is the compulsory, inseparable part of an ANN system because a well trained network performs well. After training completion, NN can be arranged to work perfectly for improved outcome.

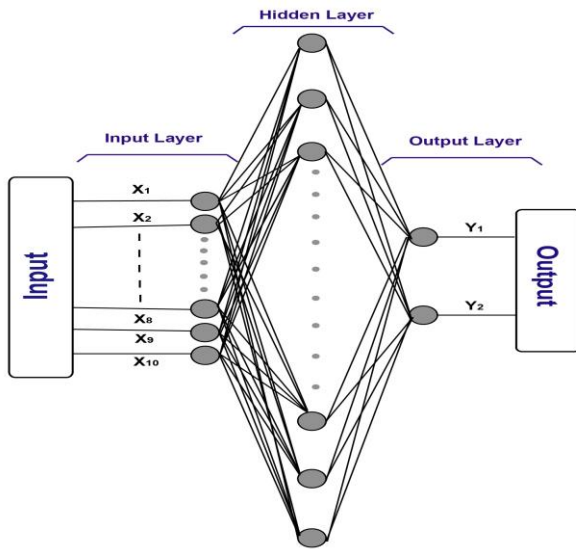


Fig. 4. LMA based NN design with 15 neuron in hidden layer.

Levenberg Marquardt Algorithm (LMA) is a fast and efficient algorithm for EEG signal classification which uses the standard numerical optimization technique. It has both property of local convergence of the Gaussian-Newton method and the global of the Gradient descent method. LMA increases the training speed, decreases the training steps as well as improves classification accuracy, though it's better to use trial and error method for better accuracy. The neural network used in this study is depicted in Fig. 4. This neural network designed with LMA where the number of neuron is chosen 15 into the hidden unit. The number of neuron is determined by doing trial and error depending on the performance accuracy of the network. 70 % of the total feature vector provided for training as good training determines outcomes well and from the rest of the data 15 % of the data used as input vector for validation and remnant 15% vectors used as testing input data.

III. RESULTS

A. Feature extraction results using DWT

Decomposition of segmented EEG signal upto level 5 is shown in Fig. 5 and in Fig. 6 for LH and RH data respectively to extract the statistical features. In both figures signal S decomposes into 6 sub-signals according to their frequencies and amplitudes. Sum of all the detail coefficients and approximation coefficient will be the equivalent of the S signal. Features for each segment data are mean, median, maximum, minimum, standard deviation, median absolute deviation, mean absolute deviation, L1 norm, L2 norm, max norm. For a segmented EEG data of left and right hand imagery movement taken from right hemisphere and left hemisphere are given in Table I.

TABLE I. EXTRACTED FEATURES OF A SEGMENTED EEG DATA FOR IMAGINATION LH/RH MOVEMENTS

Features	Left Hand	Right Hand
Mean	9263	4706
Median	9263	4706
Maximum	9275	4714
Minimum	9249	4697
Standard Dev.	4.514	2.865
Median Abs. Dev.	2.929	2.065
Mean Abs. Dev.	3.574	2.293
L1 Norm	7.123	3.619
L2 Norm	2.569	1.305

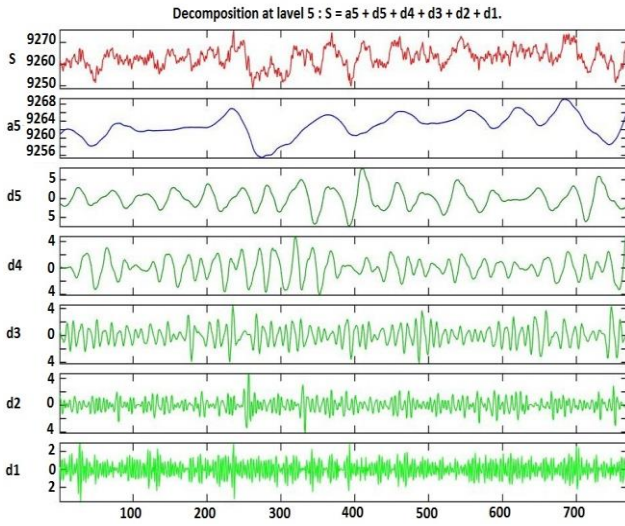


Fig. 5. Decomposition of imagery LH movement data.

B. Classification results using NN

Features collected from left and right hemisphere definitely assure differences for the imagery movement of the two hands. Now for the classification between two imagery hands movement (two classes), among 500 both hands trials, 450 trials (both hands) randomly has been used for classification. For training purposes 314 trials for training, 68 validation trials for and 68 trials for testing (both left and right) randomly used. Levenberg-Marquardt Backpropagation training algorithm used in this network because its training consumes less time. After classification confusion matrix of LMA based NN is shown in Table II. Among 100% (450) trials, for LH imagery 50% (225) trial, 42.7 % (192) trials are correctly classified. Same happens for RH trials where 49.3 % (222) trials are correctly classified from 225 trials (50%) of the total trials.

TABLE II. CONFUSION MATRIX OF NN

CONFUSION MATRIX		
Output/Desired	Left Hand Imagery Movement	Right Hand Imagery Movement
Left Hand Imagery Movement	192 42.7%	3 0.7%
Right Hand Imagery Movement	33 7.3%	222 49.3%

The performance of the classifiers was determined by the calculation of sensitivity, specificity and total classification accuracy are defined below and results shown in Table III.

$$\text{Sensitivity (True positive rate)} = \text{TP} / (\text{TP} + \text{FN}) \quad (2)$$

$$\text{Specificity (False positive rate)} = \text{TN} / (\text{TN} + \text{FP}) \quad (3)$$

$$\text{Total classification accuracy (Percentage of all samples correctly classified)} = (\text{TN} + \text{TP}) / (\text{TN} + \text{FN} + \text{TP} + \text{FP}) \quad (4)$$

Where,

Total number of correctly classified positive patterns = TP,

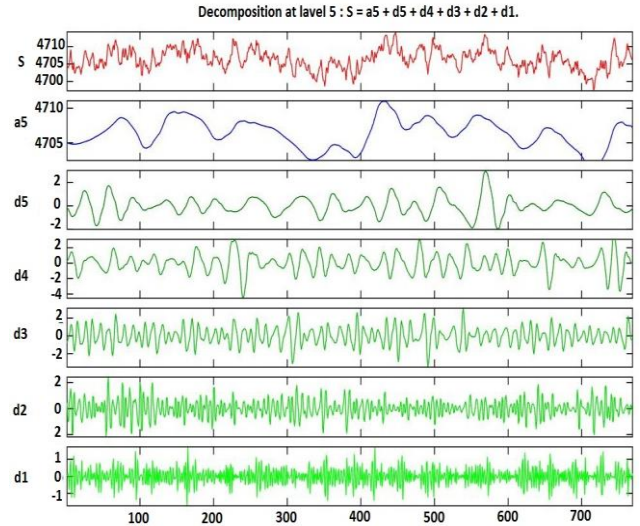


Fig. 6. Decomposition of imagery RH movement data.

Total number of actual positive patterns = TP+FN

Total number of correctly classified negative patterns = TN

Total number of actual negative patterns = TN + FP

Total number of correctly classified patterns = TN + TP

Total number of applied patterns = TN + FN + TP + FP.

TABLE III. CALCULATION OF STATISTICAL PARAMETER

Statistical Parameters		
Sensitivity (%)	Specificity (%)	Accuracy (%)
87.06	98.46	92.00

Performance curve represents the relationship among validation performance, train and test performance of a neural network with respect to the best. Validation curve determines network generalization, and to terminates training when generalization halts progressing.

The validation curves and test curves are quite analogous shown in Fig. 7. In the graph if the test curve increments meaningfully prior to the validation curve increases, possibly

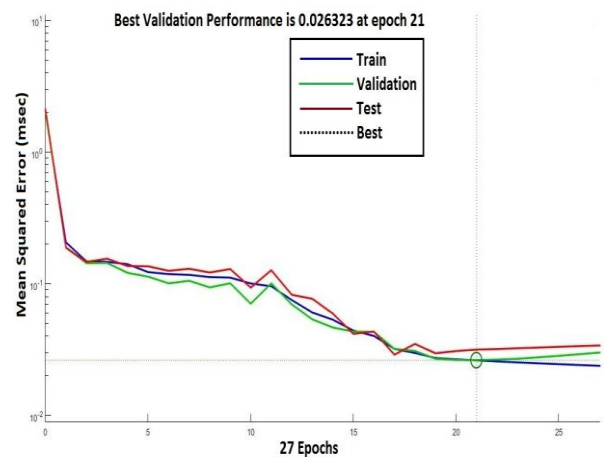


Fig. 7. Relatoin between train, validation and test curve.

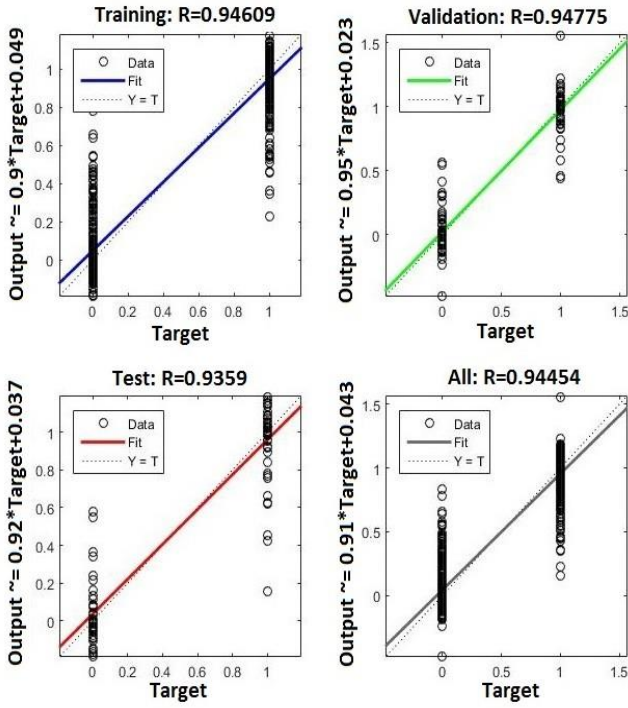


Fig. 8. Regression curve of train, validation, test and all feature vectors with 'R' value.

then some over fitting have occurred. Mean Squared Error predicts the average squared difference between outputs and targets provided in the designed NN network from the feature vectors. Lower values are acceptable and zero means no error at all. In the proposed method, validation performance is 0.026323 at epoch 21 which is shown in Fig. 7.

The four plots in Fig. 8 represent the regression, R values of training, validation, testing for all data and this pictorial view depicts the accuracy of the features vectors for train, test and validation data. The dashed line in Fig. 8 depicts the perfect result – outputs = targets. The solid line illustrates the finest fit linear regression line within outputs vectors and targets vectors. The R value demonstrates the interconnection between outputs and targets. If $R = 1$, this denotes that there is an exact linear relationship between outputs and targets feature vectors. If R is close to zero or zero, then there is no linear relationship. Here in the portrait of training, test and all data indicates that most of the data fit well. The validation results also show R values greater than 0.948. If the network performance on the validation vectors stops to increase or remains unchanged validation vectors are used to halt training early. Training ceases when the significant number of epochs are reached or the adequate amount of time is exceeded or even if the performance of network is attenuated to the target. Test vectors controls over the network generalization but do not have any influence on training.

IV. DISCUSSIONS

This proposed methodology with db4 and LMA promises satisfactory classification accuracy. Difference between the electrical activities in comparison with one hemisphere to another is expected because right half of our cerebrum controls the left side of the body and vice-versa. This assumption makes this study feasible.

We make a comparison list of some previous researches with our proposed method by comparing setup, methodology, and classification accuracy as shown in Table IV.

TABLE IV. COMPARISON OF METHODS AND CLASSIFICATION ACCURACY WITH PROPOSED METHOD

Researchers	Setup and Database	Feature Extraction And Classification Algorithms	Classification Accuracy
M. Hariharan et al. [1]	Keirn and Aunon database Seven subjects (five mental tasks).	S-transform based MSR features kNN, LDA and SVM	Between 84.72% and 98.95%
Solhjoo et al. [10]	Dataset I: EEG data set (Oxford University). Dataset2: Three channel EEG dataset (University of Technology Graz).	Mahalanobis distance, Gaussian classifiers, LDA classifiers using PSD method	63.1%, 65.4% and 65.5% respectively
M. H. Alomari et al. [11]	The EEG dataset (Physio Net). http://www.physionet.org/pn4/eegmmidb/ .	NN and SVM	89.8% and 97.1% respectively
S.-M. Zhou et al. [12]	Graz BCI data set	Higher order features for classifiers NN and LDA	90.00% and 89.29% respectively
Yue Chen et al. [13]	Graz data set B, BCI competition 2008	Levenberg-Marquardt algorithm with Multilayer feed forward neural network	78.2% and 87.1% respectively
Zhang et al. [14]	Keirn and Aunon database Four subjects with five mental tasks	High frequency power and Fischer's discriminant classifier	72.4–76.4 %
Xue et al. [15]	Six-channel, four subjects EEG data performing three mental tasks	Wavelet packet transform for feature extraction with RBF classifier	85.3 %
Proposed Method	OpenViBE / INRIA data, using the OpenViBE	Db4 wavelet and LMA based NN	92%

V. CONCLUSION

This study proved that statistical parameters (features) can be used to differentiate two imagery hands movement with standard accuracy. This proposed methodology provides convincing classification accuracy of 92 % using statistical parameters of EEG signal. In BCI or MMI application proposed method is highly applicable. In future, classification accuracy of imagery movements of hands and feet will be determined.

ACKNOWLEDGEMENT

This work was supported by Higher Education Quality Enhancement Project (HEQEP), UGC, Bangladesh; under Subproject "Postgraduate Research in BME", CP#3472, KUET, Bangladesh.

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